COMPUTER NETWORKS VIVA-VOCE QUESTION BANK

Curated Core Questions & Crisp Answers for External Viva Examination

Course Code: 2301CS501 | Covers Units 1 to 5 (Full Syllabus Revision)

UNIT 1: Introduction to Computer Networks & Physical Layer

Q1. What is a computer network and what are its key components?

A computer network is an interconnection of two or more autonomous computing devices sharing hardware, software, and data resources. Core components include end systems (hosts/clients/servers), transmission media (guided/unguided), network hardware (hubs, switches, routers), and protocols that dictate communication rules.

Q2. Distinguish between LAN, MAN, and WAN based on geographical scale and speed.

LAN (Local Area Network) covers a localized room, lab, or building (high speed, low propagation delay). **MAN (Metropolitan Area Network)** spans a town or city (e.g., cable TV/city branch network). **WAN (Wide Area Network)** connects states, countries, or continents (e.g., the global Internet, higher propagation delays).

Q3. What is the fundamental difference between Circuit Switching and Packet Switching?

In **Circuit Switching**, a dedicated end-to-end physical transmission path is reserved for the entire session duration (e.g., traditional telephony, guaranteed bandwidth, potential idle wastage). In **Packet Switching**, data is fragmented into packets routed independently via store-and-forward switches without prior reservation, maximizing link utilization.

Q4. Compare the standard Guided Transmission Media: Twisted Pair, Coaxial Cable, and Fiber Optic.

Twisted Pair: Inexpensive, wires twisted to cancel electromagnetic interference and crosstalk; used for standard Ethernet (UTP/STP). **Coaxial Cable:** Central core surrounded by dielectric and braided shield; superior noise immunity, used in cable networks. **Fiber Optic:** Glass/plastic core utilizing Total Internal Reflection of light pulses; immense bandwidth, zero EMI susceptibility, lightweight, low attenuation.

Q5. Explain the standard Physical Topologies and identify the most common LAN topology.

Topologies include **Bus** (shared single backbone; single point of failure), **Ring** (circular token loop), **Star** (central hub/switch connecting all nodes), **Mesh** (point-to-point dedicated links between every node pair; high redundancy, costly), and **Tree/Hybrid**. **Star topology** is overwhelmingly favored in modern LANs because a single node or link failure does not collapse the network.

Q6. What are the 4 fundamental components of packet delay (Total Nodal Delay)?

Total Nodal Delay = *Processing Delay* (checking bit errors & destination lookup in microseconds) + *Queuing Delay* (waiting in router buffer queue) + *Transmission Delay* (L/R, time to push packet bits onto link) + *Propagation Delay* (d/s, time for a bit to travel physical distance d at propagation speed s).

Q7. What is the difference between Transmission Delay and Propagation Delay?

Transmission delay is the time required to pump all packet bits onto the wire (L/R, dependent on packet size and link bandwidth). **Propagation delay** is the time a single bit takes to travel from sender to receiver across the physical medium (d/s, dependent on physical distance and medium velocity).

Q8. How does the 7-Layer OSI Reference Model compare to the TCP/IP Protocol Architecture?

OSI has 7 conceptual layers (Physical, Data Link, Network, Transport, Session, Presentation, Application) developed as a theoretical standard. TCP/IP is a 4-to-5 layer practical implementation model where Session and Presentation functionalities are handled directly within the Application layer.

UNIT 2: Application Layer Protocols

Q9. Contrast the Client-Server architecture with Peer-to-Peer (P2P) architecture.

In **Client-Server**, an always-on central host (server) with a static/known IP services incoming requests from intermittent clients (e.g., HTTP Web, SMTP Email). In **P2P**, arbitrary end-hosts (peers) communicate directly and distribute workloads without dedicated centralized servers, achieving high self-scalability (e.g., BitTorrent).

Q10. What is a Socket, and what parameters uniquely identify a network socket?

A socket is a software abstraction acting as the interface/doorway between an application process and the underlying transport layer. A connectionless UDP socket is identified by a 2-tuple (Destination IP, Destination Port). A connection-oriented TCP socket is uniquely bound by a 4-tuple: (Source IP, Source Port, Destination IP, Destination Port).

Q11. What is the difference between Persistent and Non-Persistent HTTP connections?

In **Non-Persistent HTTP (HTTP/1.0)**, each web object transfer requires opening a new TCP connection (incurring a 3-way handshake round-trip delay per object). In **Persistent HTTP (HTTP/1.1)**, the TCP connection remains open after transferring an object, allowing multiple consecutive requests and responses over the same socket.

Q12. What are HTTP Cookies and Web Caching (Proxy Servers)?

Cookies are state tokens set by a server in the `Set-Cookie` response header and stored by client browsers to sustain stateful interactions (e.g., shopping carts, login authentication) over stateless HTTP. **Web Caches** store recently requested web objects locally to satisfy client requests without querying origin servers, saving bandwidth and lowering latency.

Q13. How does DNS resolve a domain name into an IP address?

The Domain Name System (DNS) is a distributed hierarchical database of Root, Top-Level Domain (TLD), and Authoritative servers. When a client requests a domain, the local DNS resolver queries iteratively (or recursively) from the Root server down to the TLD and Authoritative server, returning the mapped IP (A-record) and caching it.

Q14. What are the common DNS Resource Record (RR) types?

Type A: Hostname to IPv4 mapping. **Type AAAA:** Hostname to IPv6 mapping. **Type CNAME:** Canonical name alias for a host. **Type MX:** Mail exchange server associated with the domain. **Type NS:** Authoritative name server for a zone.

Q15. Why does FTP use two separate parallel TCP connections?

FTP runs "out-of-band" using two distinct sockets: a persistent **Control Connection** on Port 21 for commands/authentication (USER, PASS, LIST) and a temporary **Data Connection** on Port 20 created and closed dynamically for transferring actual file content.

Q16. Contrast SMTP with Mail Access Protocols (POP3 and IMAP).

SMTP (Port 25) is a "push" protocol used to transfer mail messages from the sender's client to the mail server, and between mail servers. **POP3 (Port 110)** and **IMAP (Port 143)** are "pull" access protocols used by recipients to retrieve mail from their server mailbox. IMAP keeps folders synchronized on the server; POP3 typically downloads and removes from server.

UNIT 3: Transport Layer Protocols

Q17. What are the core differences between TCP and UDP?

TCP: Connection-oriented (3-way handshake), reliable, byte-stream oriented, provides flow control, congestion control, and ordered in-sequence delivery; 20-60 byte header. **UDP:** Connectionless, best-effort, lightweight (8-byte header), no guarantees for loss/ordering/congestion; ideal for real-time streaming, gaming, and DNS.

Q18. Explain the TCP 3-Way Handshake connection establishment process.

1. Client sends **SYN** segment with initial sequence number ($seq = x$).
2. Server responds with **SYN-ACK** segment ($seq = y$, $ack = x+1$).
3. Client replies with **ACK** segment ($ack = y+1$, can carry data). Connection is now established.

Q19. How does UDP detect errors if it does not guarantee delivery?

UDP includes an optional 16-bit **Internet Checksum** in its 8-byte header. The sender calculates the 1's complement sum of all 16-bit words across the segment and pseudo-header. If the receiver's computed sum contains any 0s, corruption is detected and the packet is silently discarded. Retransmission is omitted.

Q20. Differentiate between Flow Control and Congestion Control in TCP.

Flow Control: Prevents a fast sender from overwhelming a slow receiver's buffer; coordinated using the receiver's advertised window ($rwnd$) in TCP headers. **Congestion Control:** Prevents too many network sources from overwhelming intermediate router buffers across the network; governed dynamically by the sender's congestion window ($cwnd$).

Q21. Compare Go-Back-N (GBN) and Selective Repeat (SR) sliding window protocols.

In **Go-Back-N**, the receiver maintains a window of size 1, accepts only in-order packets, sends cumulative ACKs, and discards out-of-order packets. On timeout, the sender retransmits all unacknowledged packets starting from the lost one. In **Selective Repeat**, the receiver has a buffer window, ACKs packets individually, and the sender retransmits only the specific timed-out packet.

Q22. What is TCP Slow Start and how does it adjust the congestion window?

TCP Slow Start starts with a small congestion window ($cwnd = 1 ext{ MSS}$). For every received ACK, $cwnd$ increases by 1 MSS, resulting in *exponential growth* ($1 ightarrow 2 ightarrow 4 ightarrow 8 \dots$) per round-trip time until reaching a slow start threshold ($ssthresh$), after which it transitions to linear Congestion Avoidance.

Q23. What is an RPC (Remote Procedure Call)?

RPC is an inter-process communication mechanism that allows a program to execute a subroutine or function on a remote machine across a network as if it were a local function call. It hides low-level networking via client and server stubs that serialize/deserialize arguments (marshaling).

UNIT 4: Network Layer & Routing

Q24. Differentiate between Routing and Forwarding in the Network Layer.

Routing: The network-wide control-plane process of determining the best end-to-end path packets will take from source to destination using routing algorithms (OSPF, BGP). **Forwarding:** The router-local data-plane operation of moving an arriving packet from an input link interface to the appropriate output link interface using the local forwarding table.

Q25. Contrast a Datagram Network with a Virtual Circuit (VC) Network.

In a **Datagram Network** (connectionless, e.g., the Internet IP), packets are forwarded independently based on complete destination IP addresses; paths can vary dynamically. In a **Virtual Circuit Network** (connection-oriented, e.g., ATM, Frame Relay), a path is established beforehand (setup phase); packets carry VC numbers and follow the exact same path.

Q26. What is the role of the TTL (Time to Live) field in an IPv4 header?

TTL is an 8-bit counter that limits a packet's lifetime. Every intermediate router decrements TTL by 1. When TTL hits 0, the packet is discarded, and an ICMP "Time Exceeded" error is dispatched back to the sender. This prevents misrouted packets from looping indefinitely in network loops.

Q27. Explain IPv4 Classful Addressing ranges and their default subnet masks.

Class A: 1.0.0.0 – 126.255.255.255 (/8, Mask: 255.0.0.0) for very large networks.
Class B: 128.0.0.0 – 191.255.255.255 (/16, Mask: 255.255.0.0) for medium networks.
Class C: 192.0.0.0 – 223.255.255.255 (/24, Mask: 255.255.255.0) for small LANs.
Class D: 224.0.0.0 – 239.255.255.255 (Multicast). **Class E:** 240.0.0.0 – 255.255.255.255 (Experimental/Reserved). Note: 127.0.0.0/8 is reserved for Loopback testing.

Q28. What is CIDR and why is subnetting performed?

CIDR (Classless Inter-Domain Routing) eliminates rigid A/B/C classes using prefix notation (e.g., /24, /27) to prevent IP exhaustion and aggregate routing tables. **Subnetting** borrows host bits to create smaller logical broadcast domains from a single network block, improving security, traffic management, and IP utilization.

Q29. What is NAT (Network Address Translation) and why is it needed?

NAT translates private non-routable IP addresses (e.g., 10.x.x.x, 192.168.x.x) inside a local LAN to a single public IP address when accessing the Internet. It conserves IPv4 address space and adds basic network security by concealing internal host addresses from external probing.

Q30. What is ICMP and what is it used for?

Internet Control Message Protocol (ICMP) is an auxiliary network-layer protocol encapsulated within IP packets used by network devices to report errors and diagnostic information (e.g., Destination Unreachable, TTL Expired, Echo Request/Reply used by `ping` and `traceroute`).

Q31. Contrast IPv4 and IPv6 key characteristics.

IPv4: 32-bit addresses, requires manual/DHCP configuration or NAT, includes checksum and broadcast addresses. **IPv6:** 128-bit addresses (vast address space), simplified fixed 40-byte base header, eliminates checksum for speed, replaces broadcast with multicast, native auto-configuration (SLAAC), and replaces ARP with Neighbor Discovery Protocol (NDP).

Q32. Compare Distance Vector (DV) and Link State (LS) Routing Algorithms.

Distance Vector (e.g., RIP): Distributed, iterative Bellman-Ford algorithm where each router shares its full routing table only with immediate neighbors periodically; prone to routing loops and "count-to-infinity". **Link State (e.g., OSPF):** Global Dijkstra's algorithm where each router floods link states to all network nodes to build a complete topological map, computing loop-free least-cost paths.

Q33. What is Hierarchical Routing, and how do RIP, OSPF, and BGP compare?

Hierarchical routing partitions networks into autonomous systems (AS) and regions so routers only maintain details of their own local area. **RIP:** Intra-AS, Distance Vector, max 15 hops. **OSPF:** Intra-AS, Link State, fast convergence, Dijkstra-based. **BGP:** Inter-AS exterior gateway protocol using Path Vector routing across autonomous systems on the global Internet.

UNIT 5: Link Layer & Local Area Networks

Q34. What are the primary services provided by the Data Link Layer?

Framing (encapsulating network-layer datagrams with header and trailer), link access control (MAC addressing), flow control between adjacent nodes, error detection (parity, checksum, CRC), and error correction/reliable delivery over physical links.

Q35. Explain Bit Stuffing and Byte Stuffing (Character Stuffing) in framing.

Framing delimiters separate frames using special patterns (flag `01111110`). **Bit Stuffing:** Sender automatically inserts a `0` after five consecutive `1`s in the data stream so flags are not accidentally created; receiver de-stuffs the `0`. **Byte Stuffing:** Sender inserts an escape byte (`ESC`) before any accidental flag or escape byte in data; receiver strips the escape byte.

Q36. How does Cyclic Redundancy Check (CRC) error detection work?

CRC uses binary modulo-2 polynomial division (XOR operations). Sender appends $(k-1)$ zeros to data of divisor length k, divides the sequence by a predefined generator polynomial divisor, and appends the remainder (CRC checksum) to the frame. Receiver divides the incoming frame by the same divisor; a remainder of zero confirms error-free transmission.

Q37. Differentiate between Pure ALOHA and Slotted ALOHA.

Pure ALOHA: Nodes transmit immediately whenever data is ready. If collision occurs, they wait a random backoff time. Vulnerable time is $2T$; maximum efficiency is only 18.4% ($G=0.5$). **Slotted ALOHA:** Time is divided into synchronized discrete slots; nodes transmit only at slot boundaries. Vulnerable time is reduced to T, doubling efficiency to 36.8% ($G=1$).

Q38. What is Carrier Sense Multiple Access with Collision Detection (CSMA/CD)?

Used in traditional wired Ethernet. Principle: "Listen Before Transmitting & Listen While Transmitting." If the medium is sensed idle, transmit. If a collision is detected during transmission, the node immediately aborts, sends a jam signal, applies an exponential backoff algorithm, and retries.

Q39. Why do Wireless Networks (Wi-Fi) use CSMA/CA instead of CSMA/CD?

In wireless channels, collision detection is unfeasible because a transmitting radio antenna drowns out incoming weak signals (cannot hear its own collisions while transmitting) and wireless links suffer from the "hidden terminal problem". Therefore, Wi-Fi avoids collisions beforehand (CSMA/CA) using inter-frame spacing, random backoffs, and optional RTS/CTS (Request/Clear to Send) handshakes.

Q40. How does ARP (Address Resolution Protocol) resolve IP to MAC addresses?

When host A wants to send a frame to host B on the same local subnet with a known IP: Host A checks its local ARP cache. If not found, it sends an **ARP Request** (broadcast frame, MAC: `FF:FF:FF:FF:FF:FF`) asking "Who has IP X?". Host B recognizes its IP and replies with a unicast **ARP Reply** containing its physical MAC address. Host A caches the entry and sends the data.

Q41. Contrast a Layer-2 Switch with a Layer-3 Router.

Switch (Layer 2): Connects devices within the same network/LAN, inspects MAC addresses, maintains a dynamic MAC forwarding table, and isolates collision domains while remaining in a single broadcast domain. **Router (Layer 3):** Connects different networks/LANs/WANs, inspects IP addresses, uses routing tables, and breaks both collision and broadcast domains.

Q42. What is a Virtual LAN (VLAN)?

A VLAN logically segments a single physical switch into multiple isolated broadcast domains regardless of physical host locations. Hosts in different VLANs cannot communicate at Layer 2 without an intermediate Layer-3 router or multi-layer switch (inter-VLAN routing), which enhances security and curtails broadcast storms.